

Knowledge Base

Running the Smart City Demo

What this document covers

This document shows you how to run the Self-Driving Car Studio (SDCS) Smart City demo. This uses between 1-3 physical QCar 1 and 2s on either the small or large Quanser roadmap and uses 1-2 virtual QCar 2s in QLab.

Software Setup

Ensure that the following has been installed:

- Python 3.11: [Windows installer \(64-bit\)](#)
- Not Required: [Visual Studio Code](#)
- Download the Academic Resources: [Academic Resources Download](#)
- Setup the your PC to use the Quanser Python API: [Setting Up Your PC](#)
- QUARC Requirements
 - o If using QCar 1, Min Ver: QUARC 2020
 - o If using QCar 2, Min Ver: QUARC 2024 SP1
 - o This Demo was developed with QUARC 2025 (Backwards compatible)

IP Address Setup

The communications between the QCars and Ground Station rely on fixed IP addresses to be set before the demonstration is run. Please set the IP addresses as follows:

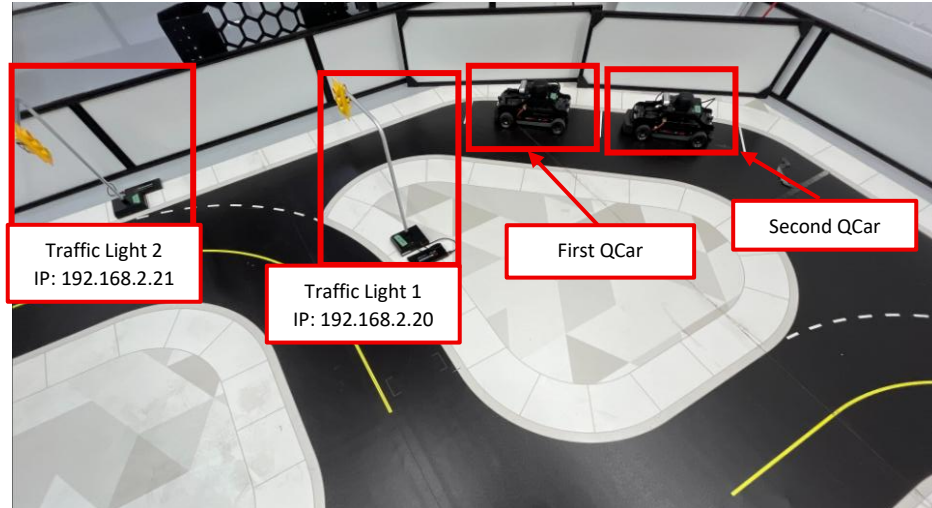
Device	IP Address
First QCar	192.168.2.11
Second QCar	192.168.2.12
Third QCar	192.168.2.13
Traffic Light 1	192.168.2.20
Traffic Light 2	192.168.2.21

As you are setting the fixed IP addresses, **do them one at a time** so that it is clear which device you are setting the IP address of. Also label the device after you have fixed the IP address.

There is another document in the **/documentation** folder called **Quanser AVRS - Setting Fixed IP Addresses v1.2.pdf** that will show you how to set static IP addresses using the router provided in the SDCS.

Hardware Setup

Place the devices in the following setup and turn them all ON:

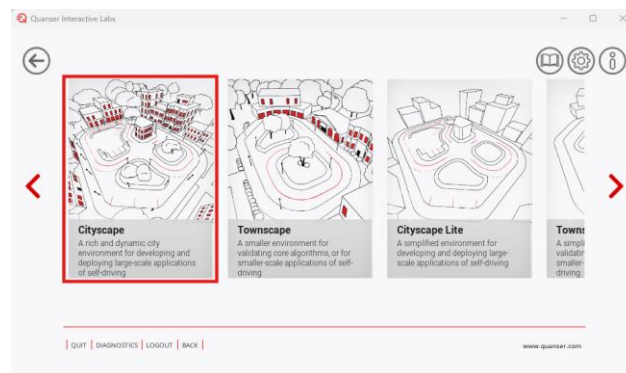


The demo will start with 1 or 2 QCars and more can be added later.

Running the Smart City Demo

Follow the below steps to run the Smart City Demo:

1. Start Quanser Interactive Labs (QLabs)
2. Navigate to the Self-Driving Car Studio, then the Cityscape (Or Cityscape Lite for better performance)



3. In a File Explorer, navigate to the **/python folder** within the Smart City Demo resources

4. Run **Setup_Smart_City.py**
 - a. The following should show in QLABs:

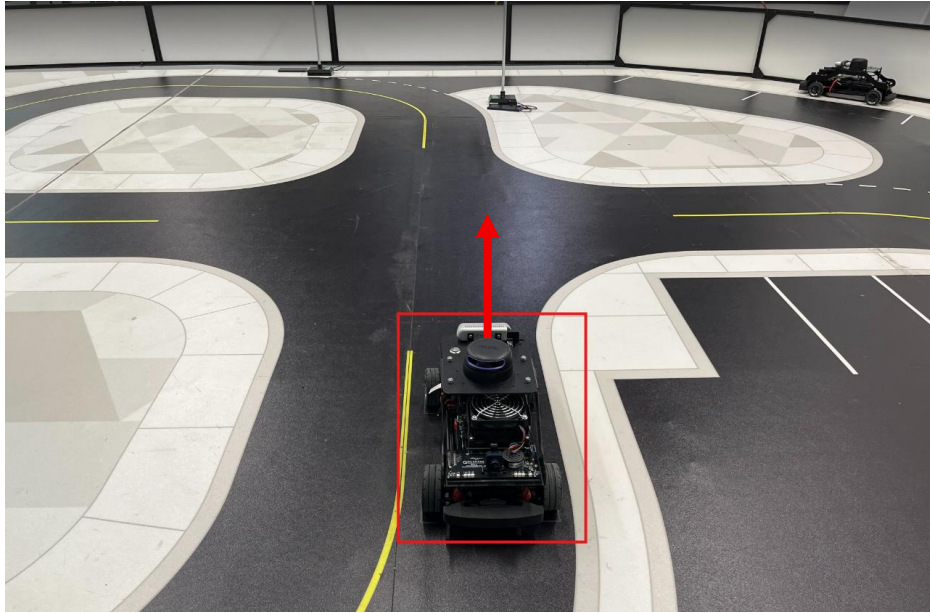


5. Run the executable **Smart_City_Communication_Infrastructure.rt-win64** by **double-clicking on it in the file explorer**
 - a. 3 QCars on the map should disappear
6. Run **Smart_City_Traffic_Controller.py**
 - a. Traffic lights should spawn in QLABs and the physical traffic lights should turn on
7. Open MATLAB and navigate to the **/Simulink folder in the MATLAB directory**

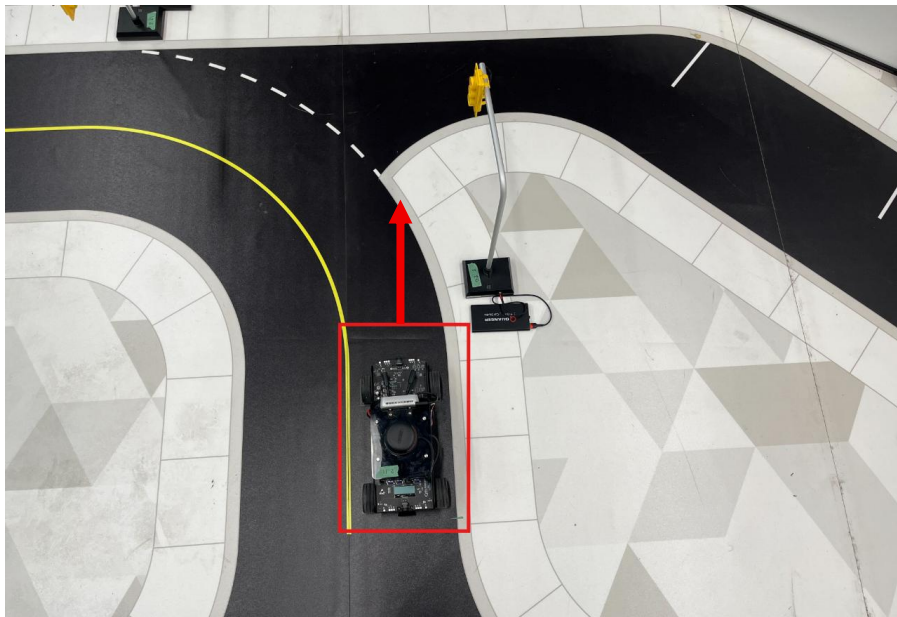
IMPORTANT FOR CALIBRATION: For the following section, you must calibrate with each type of QCar that you have (i.e. If you have both QCar 1 and 2, you must calibrate twice, once for QCar 1 and once for QCar 2)

8. Place your calibration QCar in the **location and orientation shown below (marked by a grey box on the map)** if you are using the *Large SDCS Map*

NOTE: It is important to **accurately align** the QCar within **calibration spots marked by grey boxes**

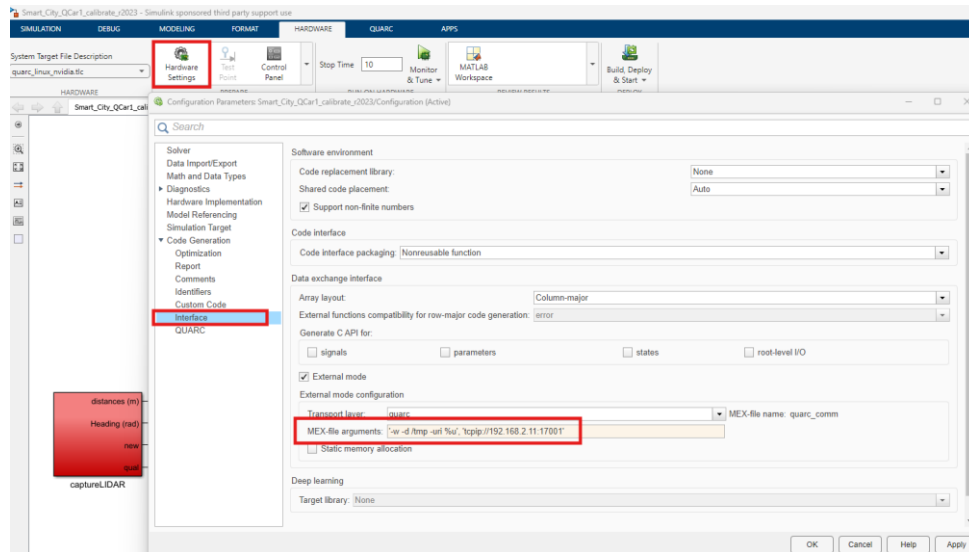


Place your QCar in the following location and orientation if you are using the *Small SDCS Map* (**also marked by a grey box**):



9. Open the correct calibration Simulink model depending on which QCar you are using to calibrate:
 - a. QCar 2: [Smart_City_QCar2_calibrate.slx](#)
 - b. QCar 1: [Smart_City_QCar1_calibrate.slx](#)

10. Ensure the IP Address in the MEX-file argument is the QCar's that you are using to calibrate:



11. Run the model using 'Monitor & Tune'
 - a. The Lidar should spin on QCar and then stop

DOUBLE-CHECK: At this point, make sure that if you have both QCar 1 and 2s, you have calibrated with each type of QCar

12. Open the [Setup_QCar1and2_params.m](#) MATLAB file
13. If you are using the Small SDCS Map, change the variable **map_type** to 0:

```
3      %% CHANGEABLE VARIABLES
4      % map_type = 1 -> Large SDCS Map
5      % map_type = 0 -> Small SDCS Map
6      map_type = 1;
7
8      % qcar_types = 3 -> VIRTUAL ONLY QCar2
9      % qcar_types = 2 -> Physical QCar 2's (+ virtual)
10     % qcar_types = 1 -> BOTH Physical QCar 1 and 2's (+ virtual)
11     % qcar_types = 0 -> Physical QCar 1's (+ virtual)
12     qcar_types = 2;
13
```

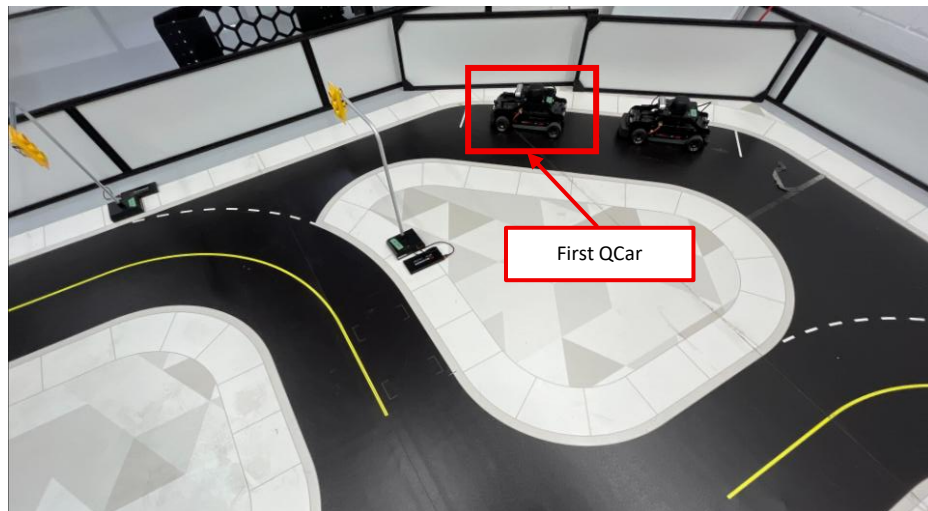
14. Change the variable **qcar_types** to match the types of QCars that you are using:

```
3 %% CHANGEABLE VARIABLES
4 % map_type = 1 -> Large SDCS Map
5 % map_type = 0 -> Small SDCS Map
6 map_type = 1;
7
8 % qcar_types = 3 -> VIRTUAL ONLY QCar2
9 % qcar_types = 2 -> Physical QCar 2's (+ virtual)
10 % qcar_types = 1 -> BOTH Physical QCar 1 and 2's (+ virtual)
11 % qcar_types = 0 -> Physical QCar 1's (+ virtual)
12 qcar_types = 2;
13
```

15. Run **Setup_QCar1and2_params.m**



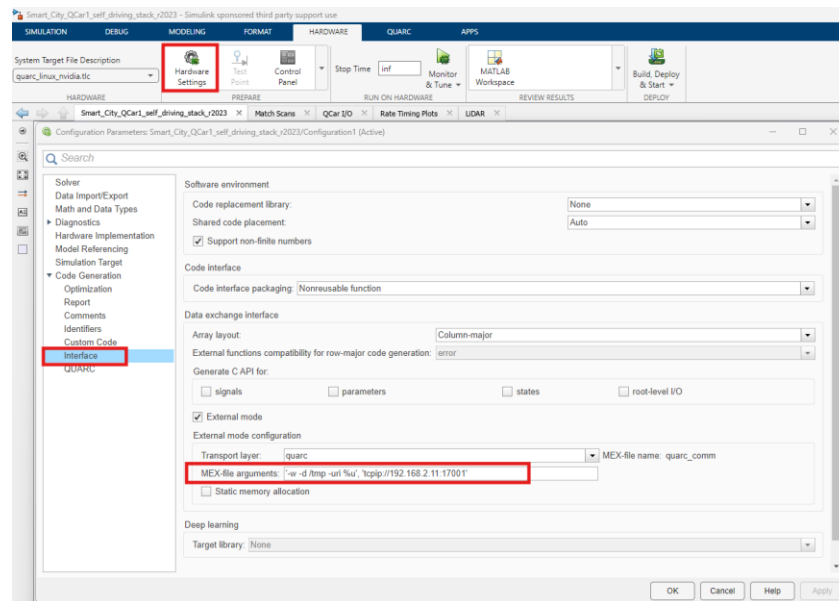
16. Return the Calibration QCar to its original position on the map



17. Open the self_driving_stack Simulink model according to whatever your **first** QCar is:

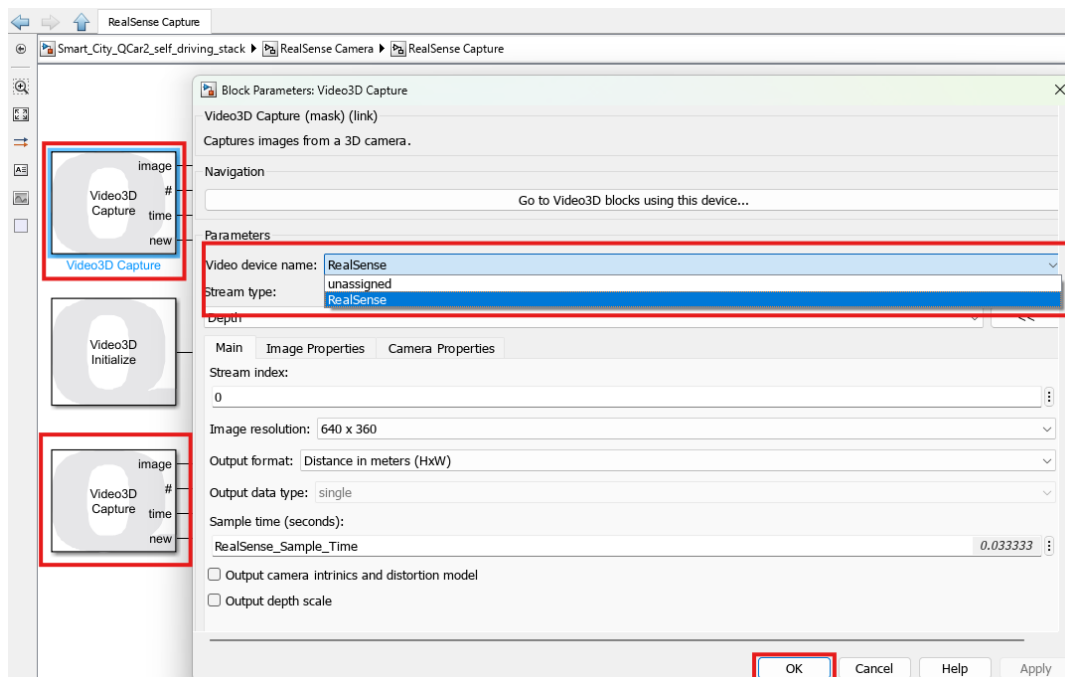
- a. QCar 2: **Smart_City_QCar2_self_driving_stack.slx**
- b. QCar 1: **Smart_City_QCar1_self_driving_stack.slx**

18. Ensure the IP Address in the MEX-file argument is the **First QCar's**:



19. Double-check Realsense camera device is assigned properly by doing the following:

- Navigate within the model to the following subsystem:
Smart_City_QCar2_self_driving_stack -> RealSense Camera -> RealSense Capture
- Ensure the *Video device name* for **BOTH Video3D Capture blocks** is set to 'Realsense':



20. Run the model using 'Monitor & Tune'



ERROR WARNING: If you receive an error like the following:

```
Error reported by S-function 'video3d_capture_block' in 'Smart_City_QCar1_self_driving_stack/RealSense_Camera/RealSense_Capture/Video3D_Capture1':  
The video device name has not been assigned. Use a Video3D Initialize block to define a video device name and then select that video device in the list of video device names.  
Component: Simulink | Category: Block diagram error
```

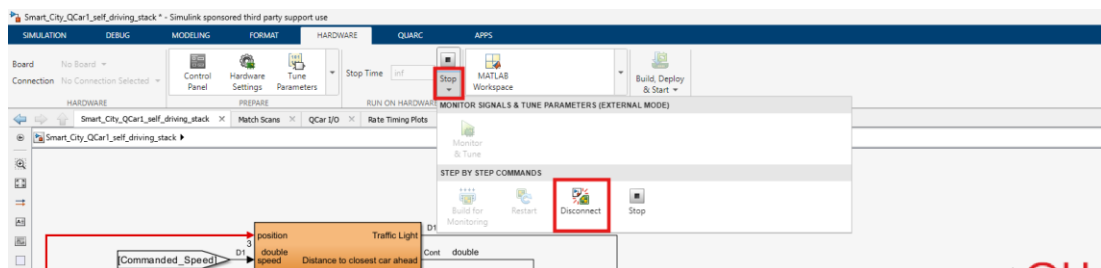
Double-Check step 19

At this point, the first QCar should be driving around the map, and the traffic lights should be switching states to allow the QCar through.

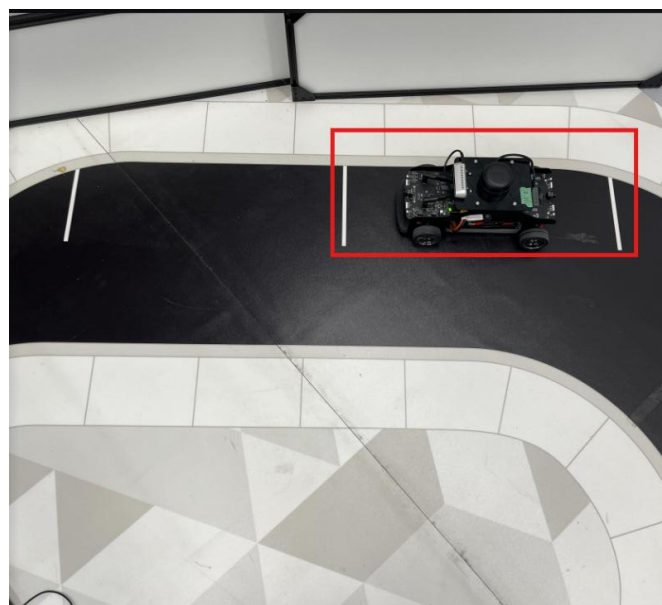
Adding Additional Physical QCars

Follow the below steps to add additional physical QCars to the city:

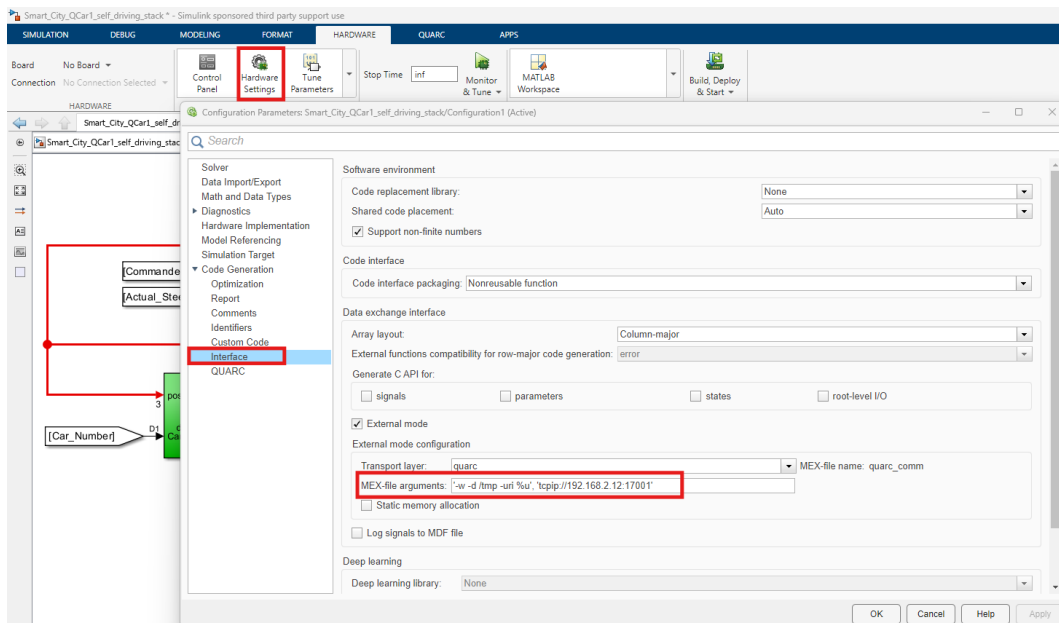
1. Within the **Self Driving Stack that is currently open** **disconnect** the Simulink model from the current QCar:



2. Place the next QCar within the parking space on the diagonal portion of the map:



3. Open the **self_driving_stack** corresponding to the QCar type you are using
4. Change the IP Address in the MEX-file argument to the next QCar's:



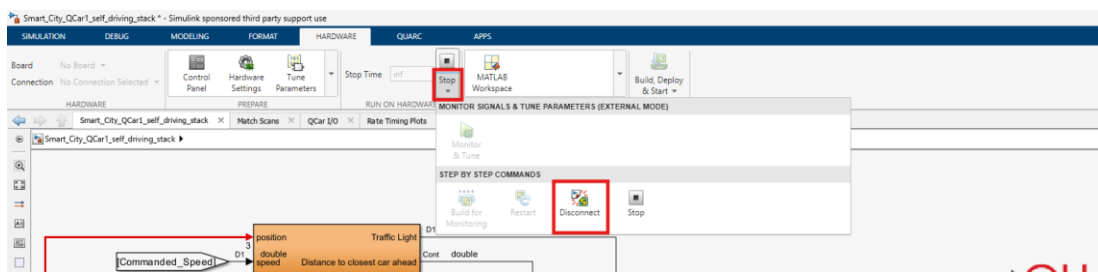
5. Run the model using 'Monitor & Tune'
6. Repeat steps 1-4 to add another QCar



Adding Virtual QCar 2s

Follow the below steps to add virtual QCar 2's to QLabS. The physical QCars will react to the virtual only ones!

1. Ensure you have **disconnected** from any Simulink models
2. Open the [Smart_City_QCar2_VIRTUAL_car4.slx](#) Simulink model
3. Run the model using 'Monitor & Tune' 
4. **Disconnect** from the [Smart_City_QCar2_VIRTUAL_car4.slx](#) Simulink model:

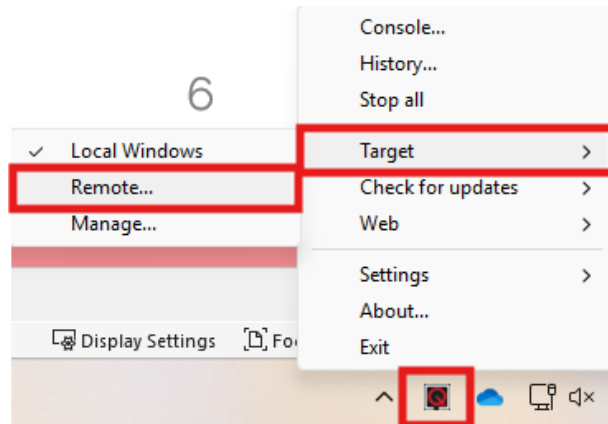


5. You may close the Simulink model once it has been run and disconnected from
6. Repeat Steps 1-3 for [Smart_City_QCar2_VIRTUAL_car5.slx](#)

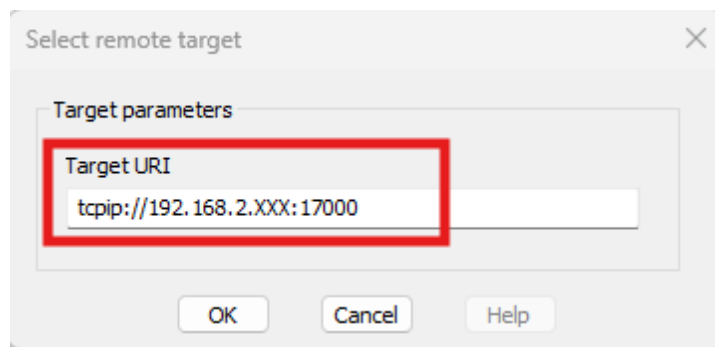
How to Stop the QCars

Do the following for each of the QCars once you finish:

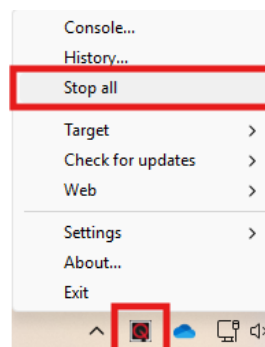
1. Open a QUARC console connected to the QCar
 - a. Click on your QUARC Icon in the Windows Icon Tray, then select 'Target' and 'Remote...'



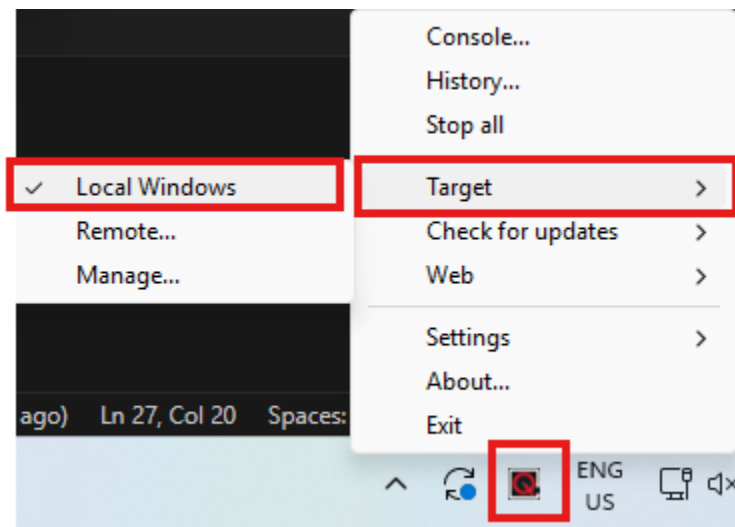
2. Input the IP ADDRESS of the QCar into the prompt (make sure you keep the tcpip:// at the beginning and the :17000 at the end)



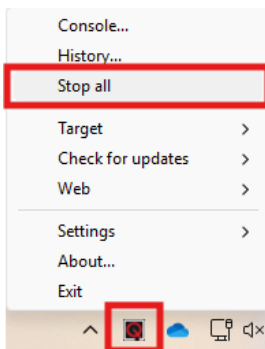
3. Select 'Stop All' from the QUARC Icon menu



4. Stop the **all local models** by selecting QUARC Icon -> Target -> Local Windows



5. Select 'Stop All' from the QUARC Icon menu



6. Stop the **Smart_City_Traffic_Controller.py** code by pressing CTRL + C in the terminal running it